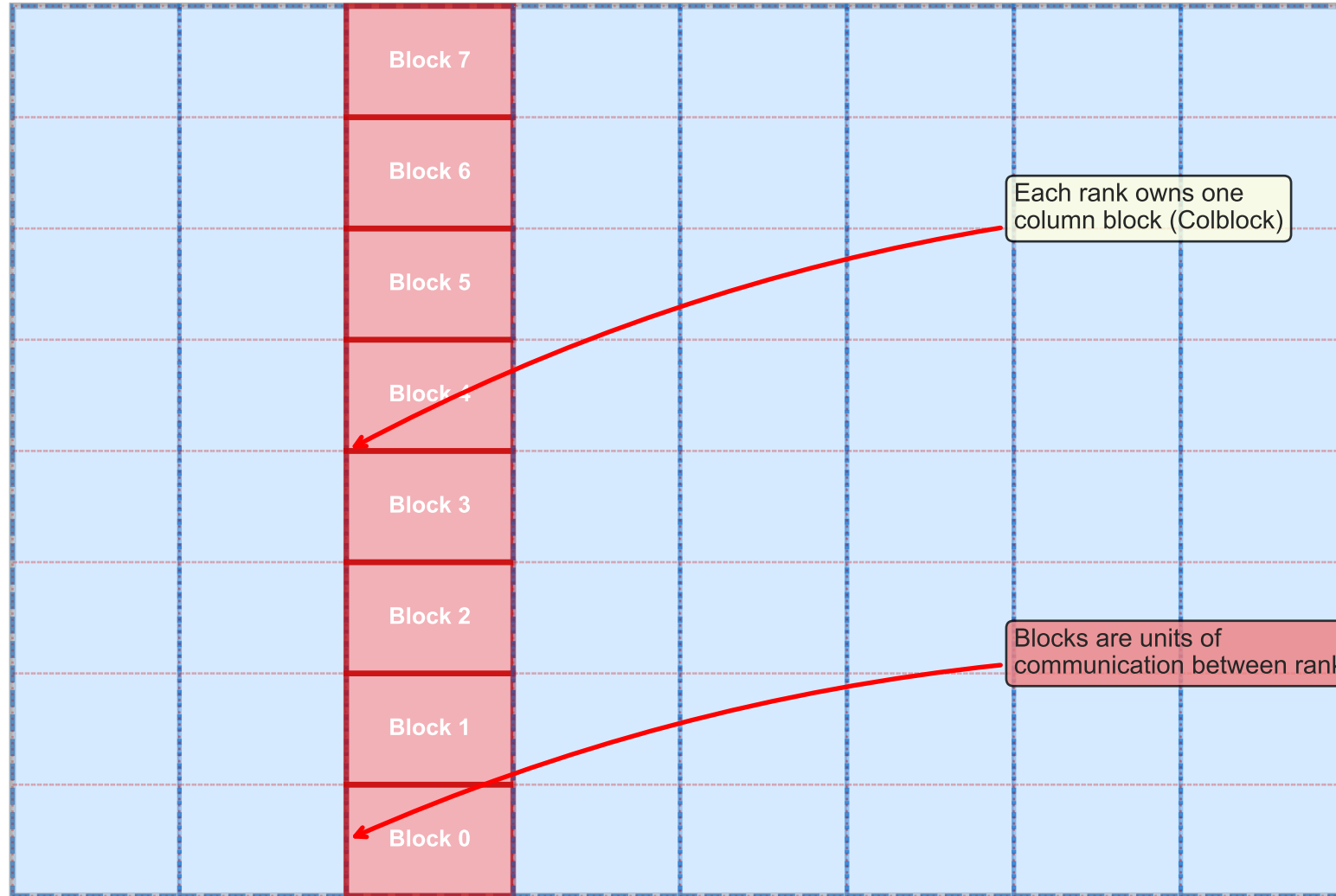


Matrix Layout for Distributed Transpose Operation

All-to-All Communication Pattern

Colblock
(One Rank)



Each rank owns one
column block (Colblock)

Blocks are units of
communication between ranks

Rank 0

Rank 1

Rank 2

Rank 3

Rank 4

Rank 5

Rank 6

Rank 7

Overall Matrix

- Each rank owns one Colblock (column block) stored contiguously in memory
- Colblocks are subdivided into Blocks for communication
- Block i of rank j is sent to rank i during transpose operation
- Storage format: column major (elements (i,j) and $(i+1,j)$ are adjacent)